Top Python Interview Questions for Data Analytics (Freshers – India)

1. What is Python and why is it used in Data Analytics?

Python is a high-level, easy-to-learn programming language used for analytics because of Pandas, NumPy, Matplotlib, Seaborn, etc.

2. What is a variable in Python?

A variable stores data, e.g., x = 10.

3. What are lists, tuples, and dictionaries?

List: Ordered, mutable

Tuple: Ordered, immutable

Dictionary: Key-value pairs

4. What is a DataFrame in Pandas?

A 2D labeled data structure like an Excel sheet.

5. How do you read a CSV file in Python?

df = pd.read_csv("file.csv")

6. Difference between loc and iloc?

loc: label-based

iloc: index-based

7. How to check missing values?

df.isnull().sum()

8. How to remove missing values?

df.dropna() or df.fillna(0)

9. What is NumPy?

Library for numerical operations.

10. List vs NumPy array?

List: slower, different data types

Array: faster, same data type

11. Merge DataFrames?

pd.merge(df1, df2, on='id')

12. Lambda function?

lambda x: x*2

13. List comprehension?

[x*2 for x in range(5)]

14. append() vs extend()?

append: adds 1 item

extend: adds multiple items

15. What is a function?

def add(a,b): return a+b

16. Filter rows?

df[df["age"] > 30]

17. Sort DataFrame?

df.sort_values("salary", ascending=False)

18. For loop?

for i in range(5): print(i)

19. Python libraries in analytics?

Pandas, NumPy, Matplotlib, Seaborn, sklearn

20. == vs is?

== compares values

is compares memory

21. Create new column?

df["new"] = df["price"] * 2

22. Drop column?

df.drop("column", axis=1)

23. Dictionary?

{"name": "John", "age": 25}

24. Handle duplicates?

df.drop_duplicates()

25. Exception handling?

try: ...

except: ...